

PROJECT REPORT

Real Estate Investment Advisor:

A Smart Data-Driven Dashboard for Property Investment Decisions

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1. Abstract

This project presents a fully functional **Real Estate Investment Advisor** — an interactive Streamlit dashboard that helps potential investors make smarter decisions in the Indian real estate market. Using a dataset of 250,000+ properties, the application performs data cleaning, feature engineering, custom ranking, and comprehensive EDA. Users can filter properties by city, BHK, price range, premium segment, and investment score, while receiving live business insights on every chart.

The solution eliminates guesswork by combining rule-based Good Investment logic with a custom Value_Rank_Score based on amenities, size, parking, security, schools, hospitals, and price. The dashboard is clean, responsive, and production-ready.

2. Introduction

Real estate is one of the largest investment sectors in India, yet most individual investors make decisions based on limited information. This project was developed to bridge that gap by providing a powerful, user-friendly data analytics tool that delivers actionable insights at the click of a button.

3. Business Problem & Objective

Business Problem People invest in real estate without proper analysis. They often don't know:

- Which property offers the best value
- Which location or features truly increase returns
- How to compare thousands of options quickly

Business Objective

- Help users identify **good properties** to invest in
- Help users understand **factors** affecting property prices
- Enable **smart, data-driven** real estate decisions

4. Dataset Description

- **Name:** india_housing_prices.csv
- **Rows:** 250,000

- **Columns:** 23
- **Key Features:** City, Locality, Property_Type, BHK, Size_in_SqFt, Price_in_Lakhs, Price_per_SqFt, Amenities, Parking_Space, Security, Nearby_Schools, Nearby_Hospitals, Public_Transport_Accessibility, Owner_Type, Availability_Status, etc.
- **Quality:** Zero missing values, zero duplicates after cleaning — exceptionally clean dataset.

5. Methodology

The project follows a structured data analytics pipeline:

1. Data Understanding & Cleaning
2. Feature Engineering
3. Custom Ranking & Rule-based Logic
4. Exploratory Data Analysis (20+ visualizations)
5. Interactive Streamlit Dashboard Development

No machine learning models were used — the solution relies on transparent, rule-based logic and a custom scoring system as per project requirements.

6. Data Preprocessing & Feature Engineering

- Handled missing values and data types
- Created Age_of_Property and Price_per_SqFt
- Generated Amenity_Count and Premium_Segment (Low / Medium / High)
- Created Security_Score
- Built **Value_Rank_Score** using your exact criteria: $\text{Amenity_Count} \times 2 + \text{Size}/1000 + \text{Parking} \times 5 + \text{Security} \times 10 + \text{Price}/50 + \text{Schools} \times 8 + \text{Hospitals} \times 8$

7. Exploratory Data Analysis (EDA)

All 20 EDA points specified in the project flow were implemented with interactive visualizations and live business insights:

- Price & Size analysis (distributions, outliers, relationship)
- Location-based analysis (State, City, Locality)
- Relationships & Correlation (including clean, readable heatmap)
- Amenities, Parking, Transport, Schools, Hospitals impact
- Owner Type and Availability Status impact

8. Good Investment Logic & Value Ranking System

Good Investment Rule (as per project flow): Low Price/SqFt + High Amenities + High Transport + Good Schools + Newer properties.

Value_Rank_Score (custom ranking you requested): Prioritizes higher amenity count, larger size, more parking, better security, more schools/hospitals, and premium pricing.

9. Streamlit Dashboard Features

- Sidebar filters: City, BHK, Price range, Premium Segment, Good Investments only
- Flexible sorting options (Value Rank Score is default)
- Dynamic business insights under every graph
- Premiumness segmentation view
- Clean, professional UI with download option
- Fully responsive and user-friendly

10. Key Insights & Business Recommendations

Major Insights:

- Size and Price have strong positive correlation (~0.7)
- High Premium properties (rich amenities) consistently show higher returns
- Location remains the #1 price driver
- Schools and hospitals add clear value premium
- Newer properties appreciate faster

Business Recommendations:

- Target **High Premium** properties with high Value_Rank_Score for maximum long-term appreciation
- Prioritize properties with strong school + hospital + transport access for family buyers
- Use the dashboard's ranking system to quickly identify best value opportunities
- Mid-tier cities offer excellent risk-adjusted returns compared to metros

11. Conclusion & Future Scope

This project successfully delivers a complete, end-to-end real estate investment advisor that meets all requirements of the project flow document. The dashboard is practical, insightful, and ready for real-world use.

Future Scope:

- Add future price prediction using ML
- Deploy on Streamlit Cloud for public access
- Add PDF report export feature

12. References

- Project Flow Document (real-estate-project-flow.pdf)
- Project Title & Guidelines Document
- Streamlit Documentation
- Pandas & Plotly Documentation